D. Points

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Pete and Bob invented a new interesting game. Bob takes a sheet of paper and locates a Cartesian coordinate system on it as follows: point $(0, 0)$ is located in the bottom-left corner, Ox axis is directed right, Oy axis is directed up. Pete gives Bob requests of three types:

- `add x y` — on the sheet of paper Bob marks a point with coordinates (x, y). For each request of this type it's guaranteed that point (x, y) is not yet marked on Bob's sheet at the time of the request.
- `remove x y` — on the sheet of paper Bob erases the previously marked point with coordinates (x, y). For each request of this type it's guaranteed that point (x, y) is already marked on Bob's sheet at the time of the request.
- `find x y` — on the sheet of paper Bob finds all the marked points, lying strictly above and strictly to the right of point (x, y). Among these points Bob chooses the leftmost one, if it is not unique, he chooses the bottommost one, and gives its coordinates to Pete.

Bob managed to answer the requests, when they were 10, 100 or 1000, but when their amount grew up to $2 \cdot 10^5$, Bob failed to cope. Now he needs a program that will answer all Pete's requests. Help Bob, please!

Input

The first input line contains number n $(1 \le n \le 2 \cdot 10^5)$ — amount of requests. Then there follow n lines — descriptions of the requests. `add x y` describes the request to add a point, `remove x y` — the request to erase a point, `find x y` — the request to find the bottom-left point. All the coordinates in the input file are non-negative and don't exceed 10^9.

Output

For each request of type `find x y` output in a separate line the answer to it — coordinates of the bottommost among the leftmost marked points, lying strictly above and to the right of point (x, y). If there are no points strictly above and to the right of point (x, y), output `-1`.

Examples

input Copy

```
7
add 1 1
add 3 4
find 0 0
remove 1 1
find 0 0
add 1 1
find 0 0
```

output Copy

```
1 1
3 4
1 1
```

input Copy

```
13
add 5 5
add 5 6
add 5 7
add 6 5
add 6 6
add 6 7
add 7 5
add 7 6
add 7 7
find 6 6
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 19

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371206374	Apr/15/2026 08:42	Accepted
371206237	Apr/15/2026 08:40	Compilation error
371206064	Apr/15/2026 08:38	Wrong answer on test 3
371205898	Apr/15/2026 08:35	Time limit exceeded on test 30

```
remove 7 7
find 6 6
find 4 4
```

output Copy

```
7 7
-1
5 5
```

→ Problem tags

data structures *2800

No tag edit access

→ Contest materials

- Announcement
- Tutorial (en)

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The only programming contests Web 2.0 platform
Server time: Apr/15/2026 12:48:01 UTC+7 (n2).
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